KnowForge AI

Industrial Knowledge Intelligence Platform

ET AI Hackathon 2.0 – Phase 2

Problem Statement:

PS 8 – AI for Industrial Knowledge Intelligence:

Unified Asset & Operations Brain

Team Name:

VIKLON

Team Members:

VIKAS CHAVAN

Date: 22/07/2026

Executive Summary

KnowForge AI is an AI-powered Industrial Knowledge Intelligence Platform developed to help organizations efficiently search, retrieve, and understand information from large collections of industrial documents.

Instead of manually searching through hundreds of PDF manuals, SOPs, maintenance guides, inspection reports, and technical documents, users can upload their documents into the platform. The system automatically extracts the content, creates semantic knowledge embeddings, and builds an intelligent searchable knowledge base.

Using Retrieval-Augmented Generation (RAG), users can ask natural language questions and receive context-aware answers along with the exact source document and page reference.

This significantly reduces document search time, improves operational efficiency, and enables faster decision-making across industrial environments.

Problem Statement

Industrial organizations maintain thousands of technical documents such as SOPs, operation manuals, maintenance logs, inspection reports, safety guidelines, and engineering documents.

Finding relevant information within these documents is often slow, manual, and inefficient.

Traditional keyword search fails to understand context and relationships between information, causing engineers and operators to spend valuable time locating required knowledge.

The challenge is to develop an intelligent AI-powered knowledge platform capable of understanding industrial documents and providing accurate contextual answers with traceable sources.

Proposed Solution

KnowForge AI provides a Retrieval-Augmented Generation (RAG) platform for industrial knowledge management.

The system allows organizations to upload PDF documents which are automatically processed through text extraction, semantic chunking, vector embedding generation, and FAISS indexing.

When users ask questions, the system retrieves the most relevant knowledge chunks using semantic similarity search and provides accurate responses using Google's Gemini language model.

Each response includes source references and page numbers, ensuring transparency and explainability.

Objectives

• Build an intelligent industrial document search platform.

• Reduce document retrieval time.

• Provide context-aware AI responses.

• Support semantic search instead of keyword search.

• Display source references for every answer.

• Create a scalable industrial knowledge base.

System Architecture

KnowForge AI follows a Retrieval-Augmented Generation (RAG) architecture for intelligent industrial knowledge retrieval. Uploaded PDF documents are processed through an AI pipeline consisting of text extraction, semantic chunking, embedding generation, vector indexing using FAISS, semantic retrieval, and Gemini-powered answer generation. Every response is accompanied by the original document and page reference, ensuring explainability and traceability.

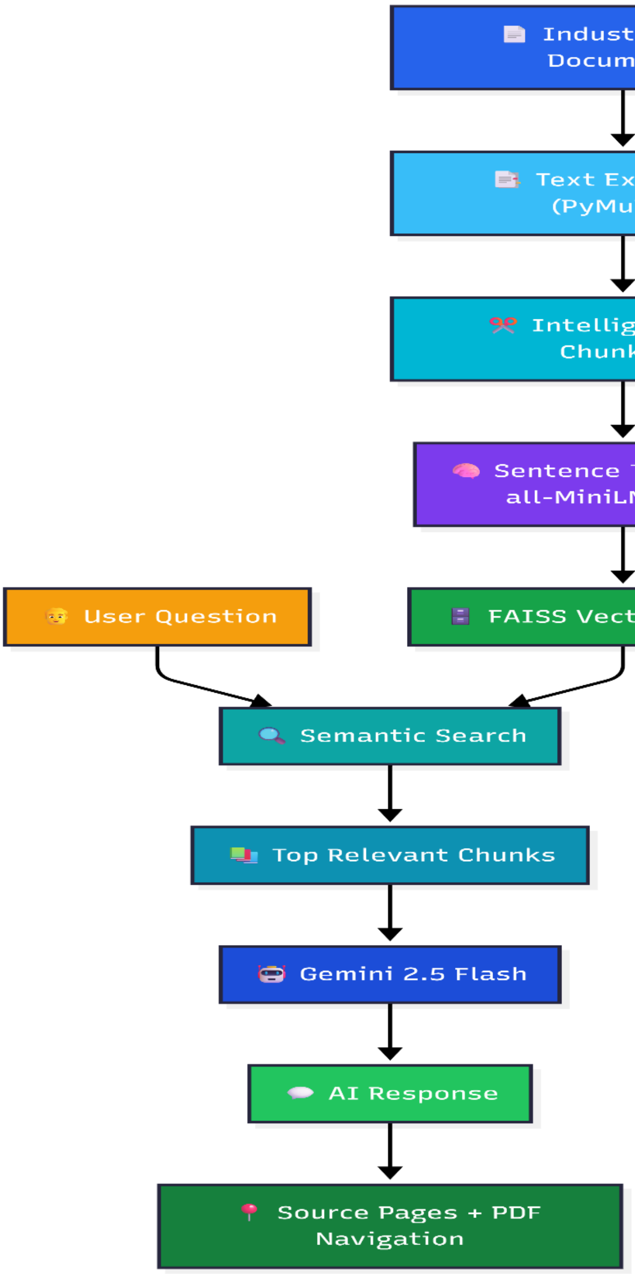


Figure 1: Overall Retrieval-Augmented Generation (RAG) Architecture of KnowForge AI.

Workflow Explanation

1. Document Upload

Users upload industrial PDF documents such as SOPs, maintenance manuals, inspection reports, and operational guides.

2. Text Extraction

The system extracts structured text from uploaded PDF documents using PyMuPDF.

3. Semantic Chunking

Large documents are divided into smaller context-preserving chunks to improve retrieval quality.

4. Embedding Generation

Each chunk is converted into high-dimensional semantic vectors using the Sentence Transformer model (all-MiniLM-L6-v2).

5. Knowledge Base Construction

The generated embeddings are indexed into a FAISS vector database, enabling fast semantic search.

6. User Query Processing

When a user submits a natural language query, it is converted into an embedding using the same embedding model.

7. Semantic Retrieval

FAISS identifies the most relevant knowledge chunks using cosine similarity and retrieves the top matching results.

8. AI Answer Generation

The retrieved context is provided to Gemini 2.5 Flash, which generates a context-aware response without relying on unrelated external knowledge.

9. Explainable Results

The final answer is displayed together with document names, similarity scores, page numbers, and direct PDF page navigation for verification.

Technology Stack

Frontend

• React

• Vite

• TailwindCSS

• Axios

Backend

• FastAPI

• Python

AI

• Google Gemini 2.5 Flash

• Sentence Transformers

• all-MiniLM-L6-v2

Vector Database

• FAISS

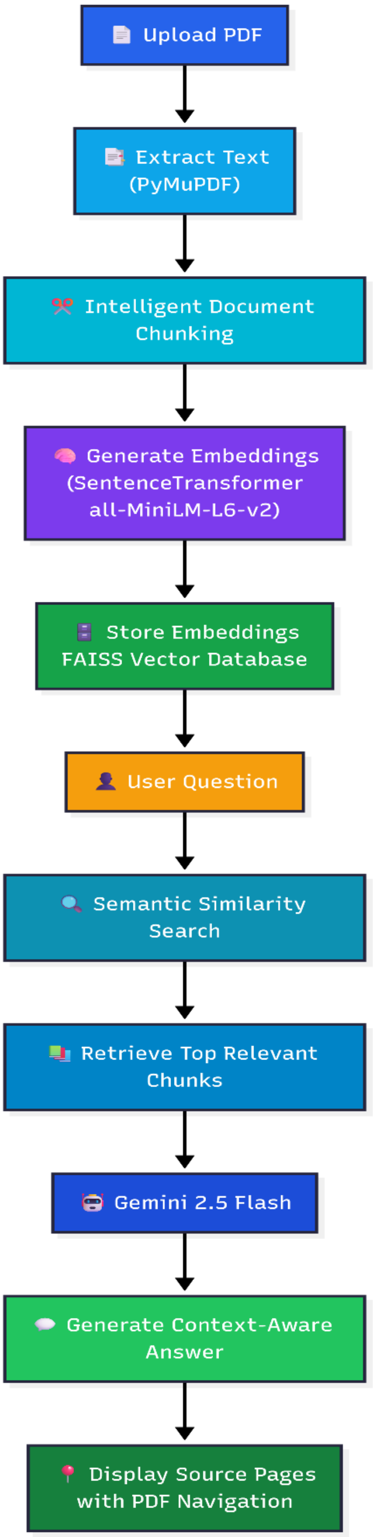
Document Processing

• PyMuPDF

PDF Viewer

• React-PDF

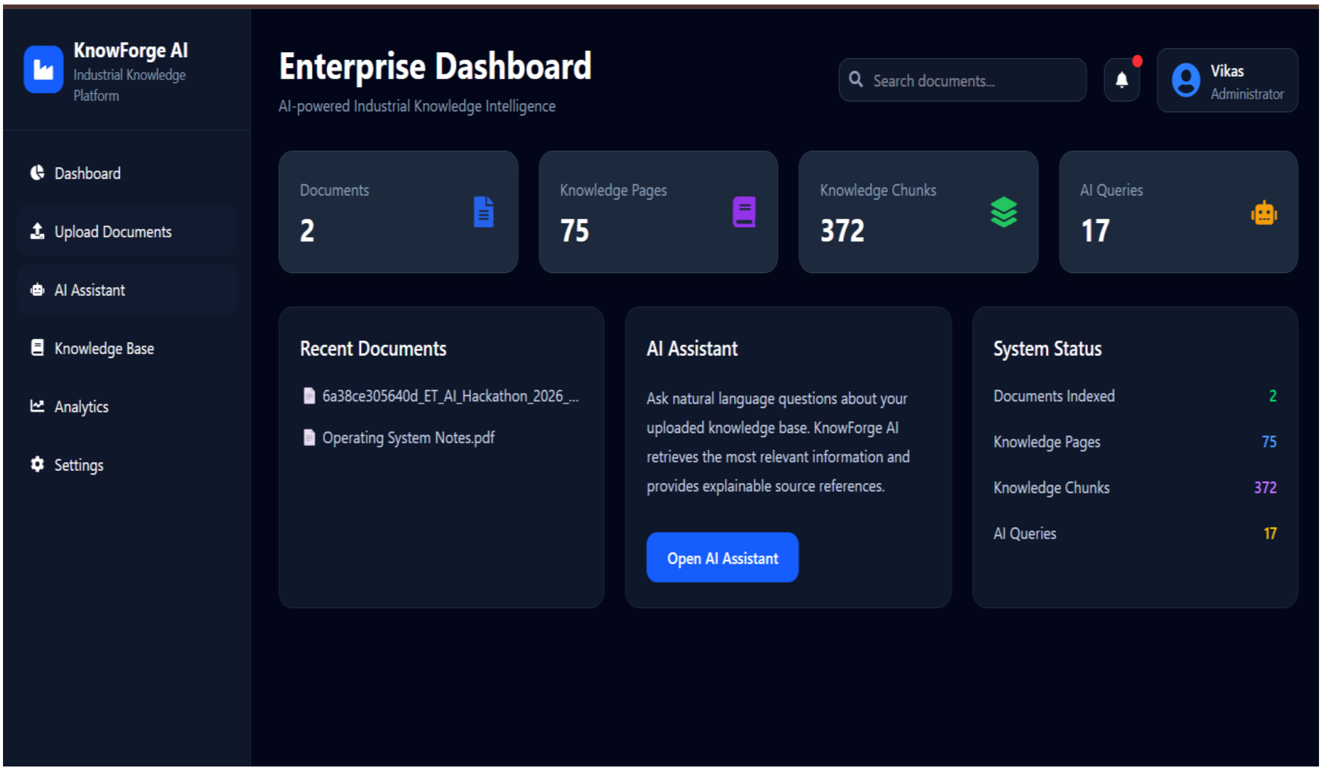
Working Flow

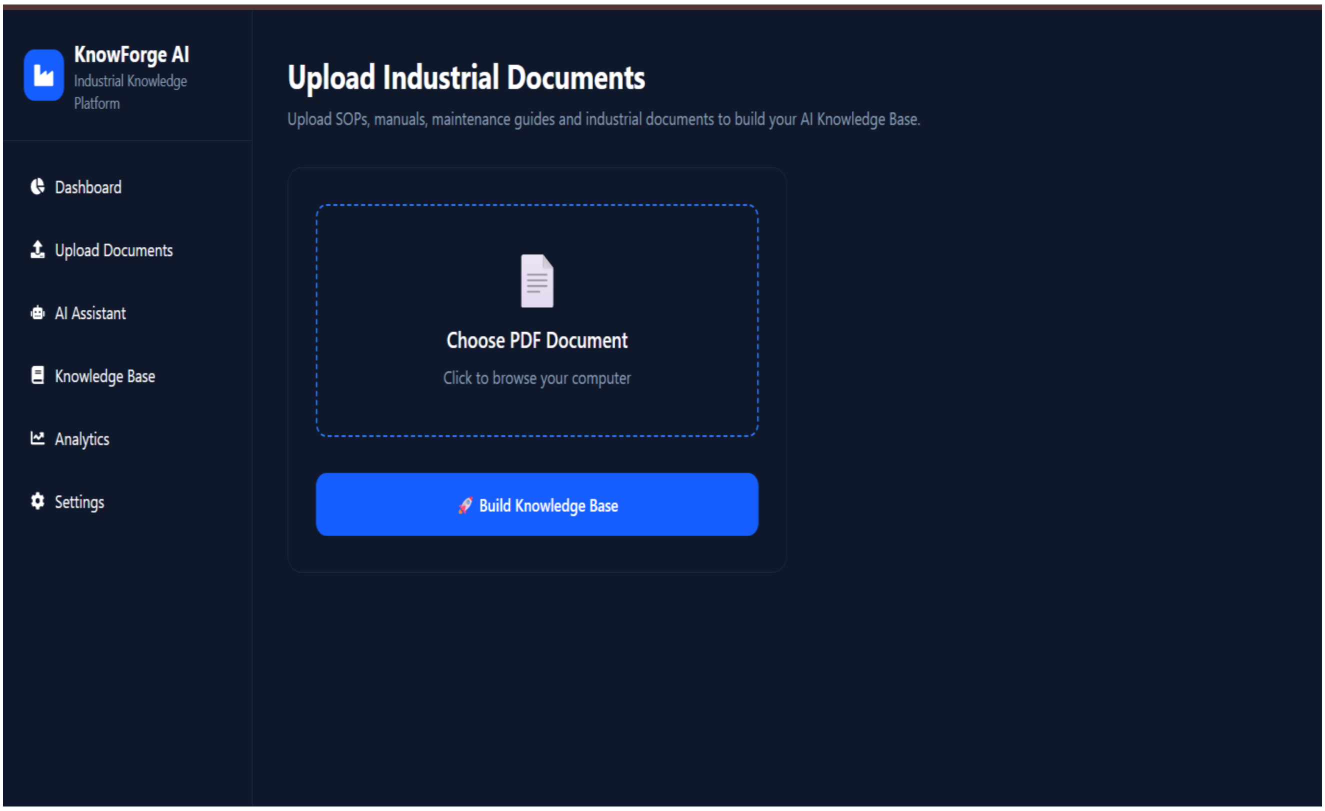


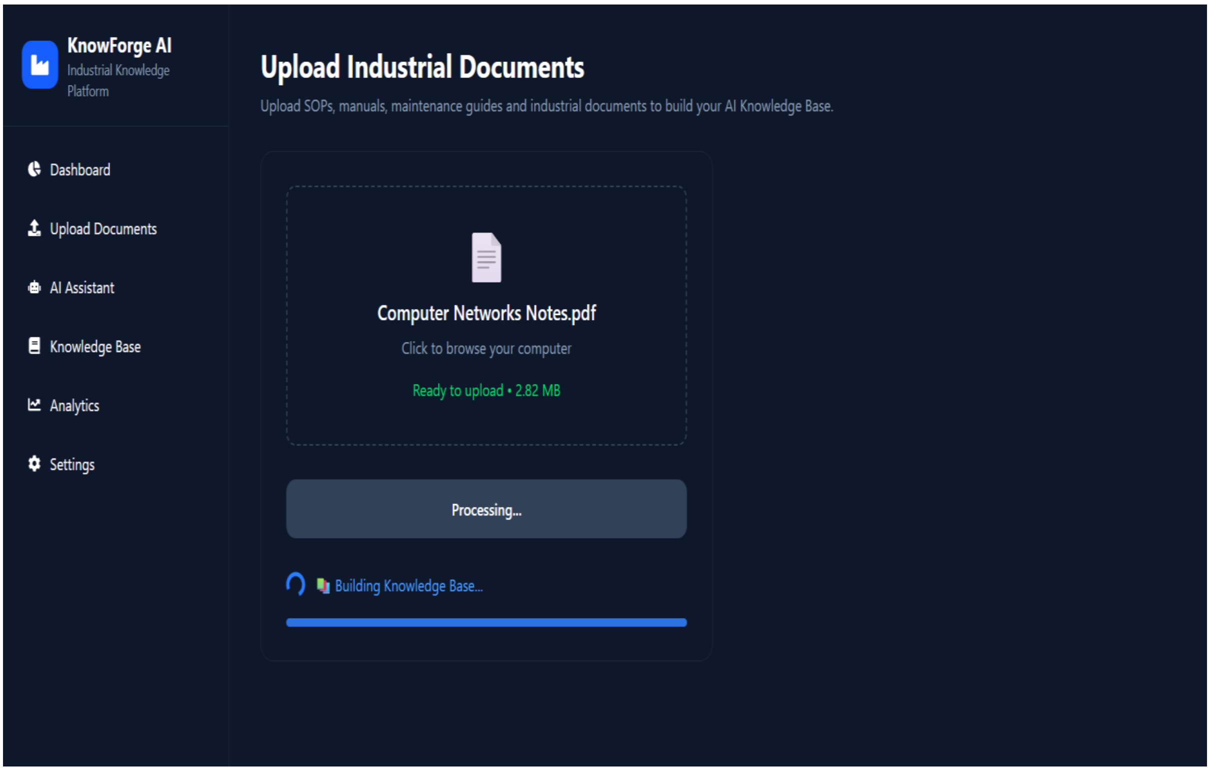
Features

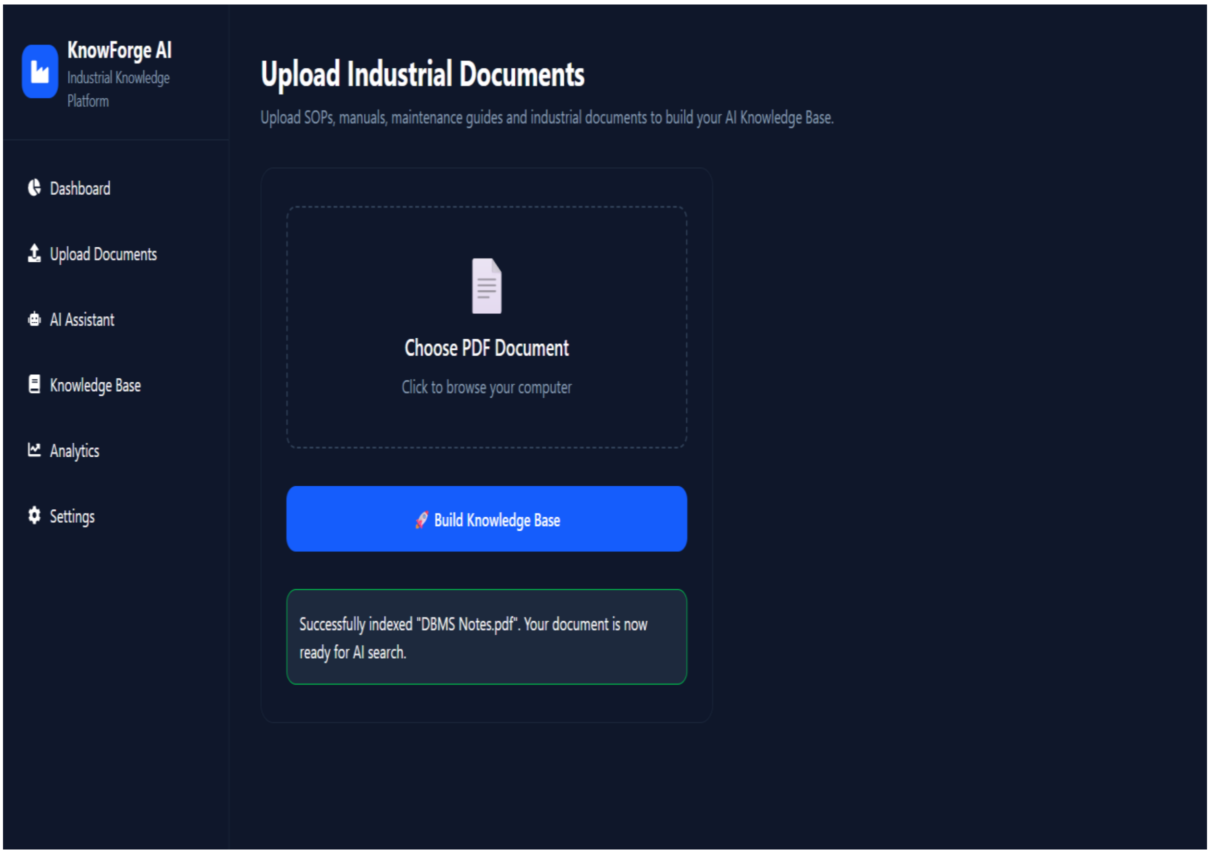
- Upload Industrial PDFs
- Automatic Text Extraction
- Intelligent Chunking
- AI Question Answering
- Semantic Search
- Source Page References
- PDF Preview
- Knowledge Base Dashboard
- Document Statistics
- Analytics Dashboard
- Delete Documents
- Real-time Knowledge Indexing

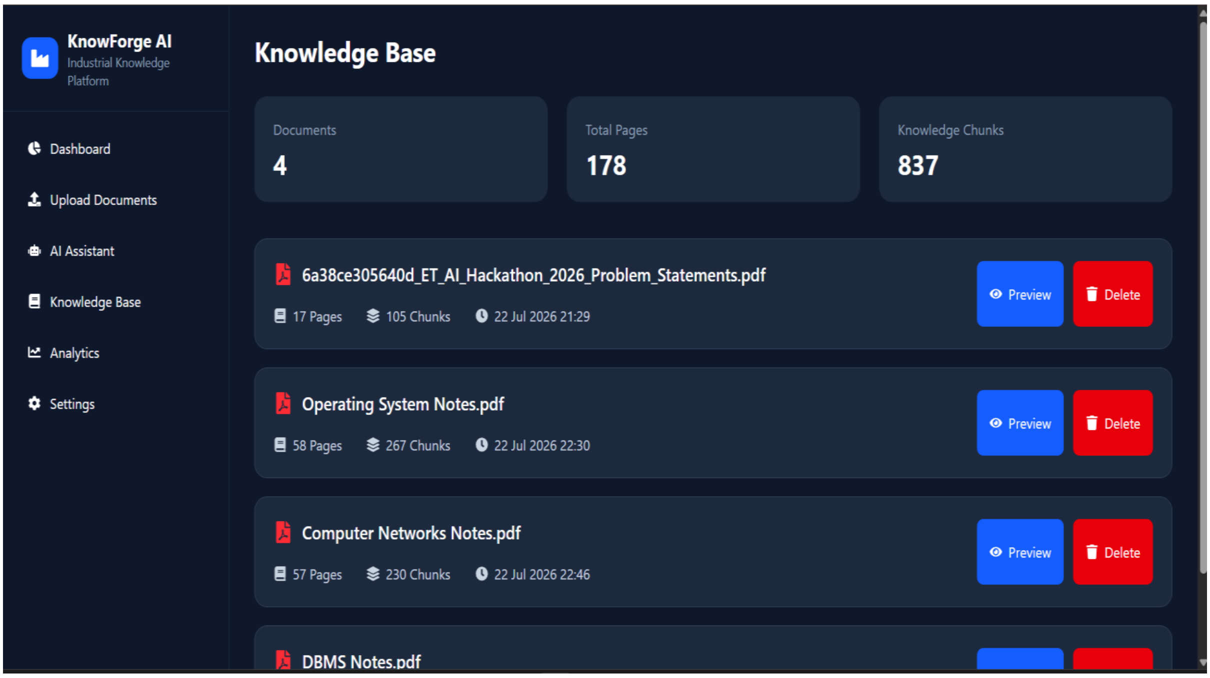
Screenshots






KnowForge AI
Industrial Knowledge Platform
Dashboard
Upload Documents
AI Assistant
Knowledge Base
Analytics
Settings
Upload Industrial Documents
Upload SOPs, manuals, maintenance guides and industrial documents to build your AI Knowledge Base.
Computer Networks Notes.pdf
Click to browse your computer
Ready to upload • 2.82 MB
Processing...
Building Knowledge Base...


KnowForge AI
Industrial Knowledge Platform
Dashboard
Upload Documents
AI Assistant
Knowledge Base
Analytics
Settings
Upload Industrial Documents
Upload SOPs, manuals, maintenance guides and industrial documents to build your AI Knowledge Base.
Choose PDF Document
Click to browse your computer
Build Knowledge Base
Successfully indexed "DBMS Notes.pdf". Your document is now ready for AI search.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Industrial Knowledge Platform
Dashboard
Upload Documents
AI Assistant
Knowledge Base
Analytics
Settings
Knowledge Base
Documents
4
Total Pages
178
Knowledge Chunks
837
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17 Pages
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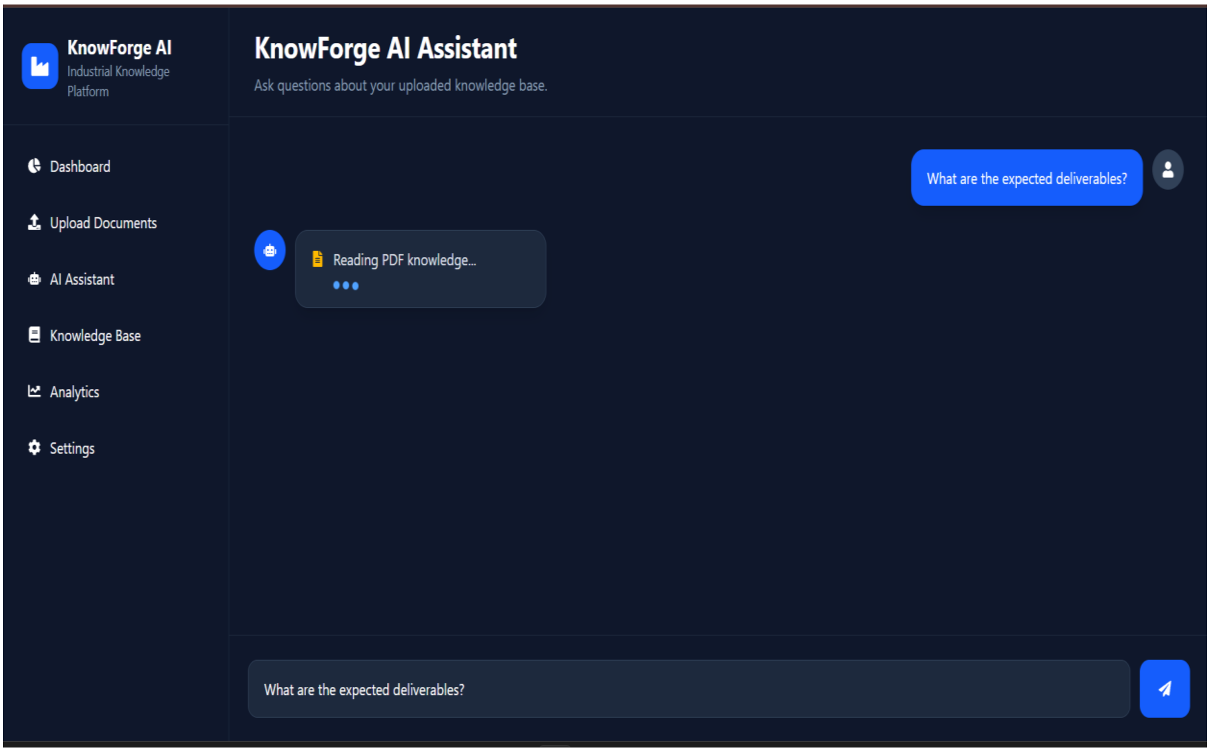
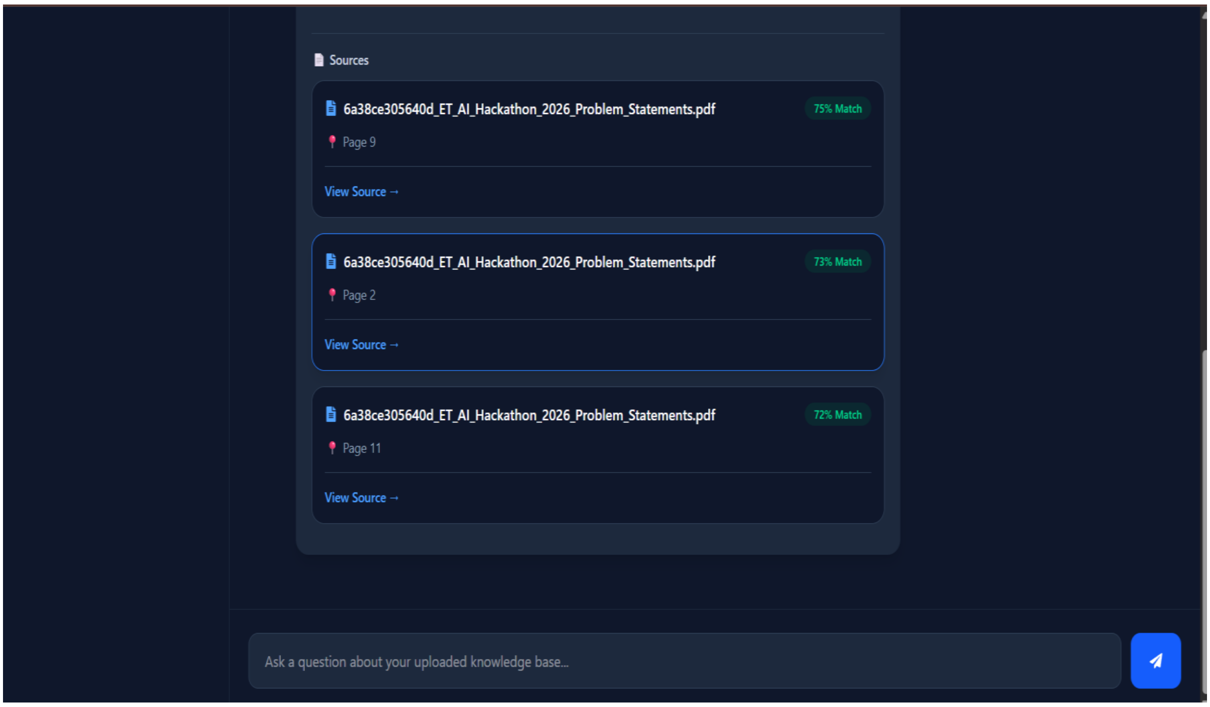

KnowForge AI
Industrial Knowledge Platform
Dashboard
Upload Documents
AI Assistant
Knowledge Base
Analytics
Settings
KnowForge AI Assistant
Ask questions about your uploaded knowledge base.
What are the expected deliverables?
Reading PDF knowledge...
What are the expected deliverables?


Sources
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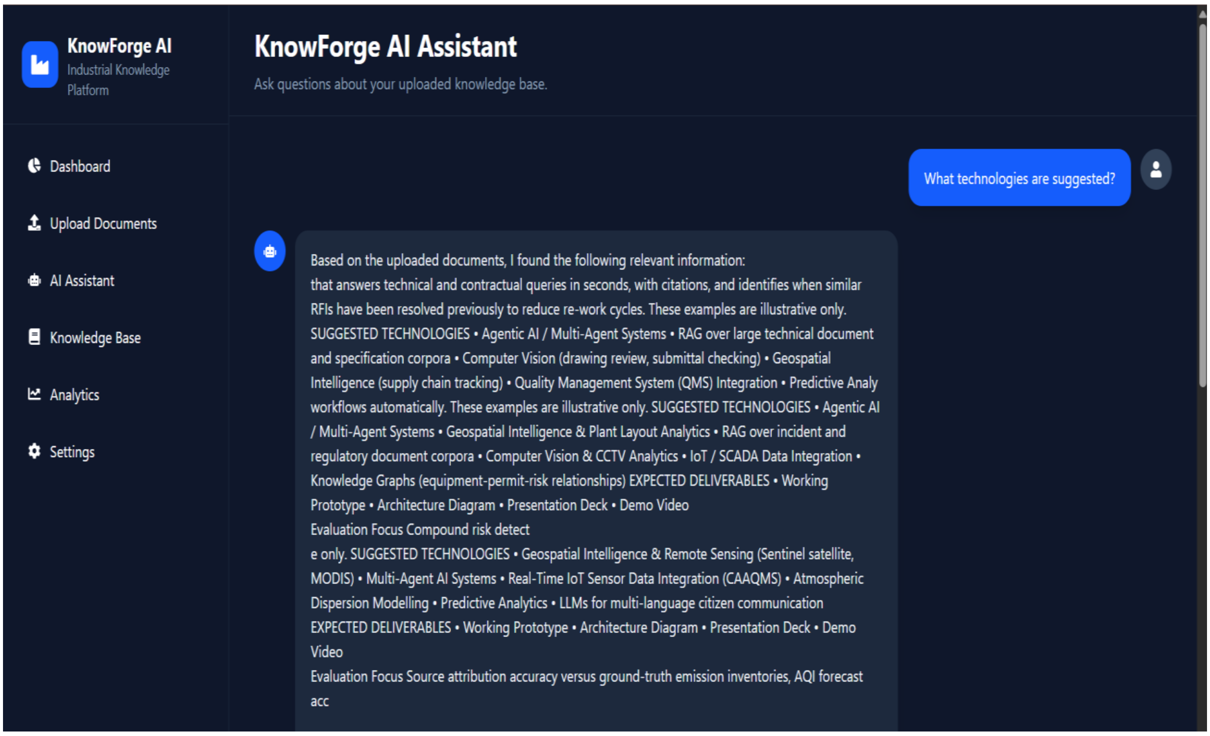
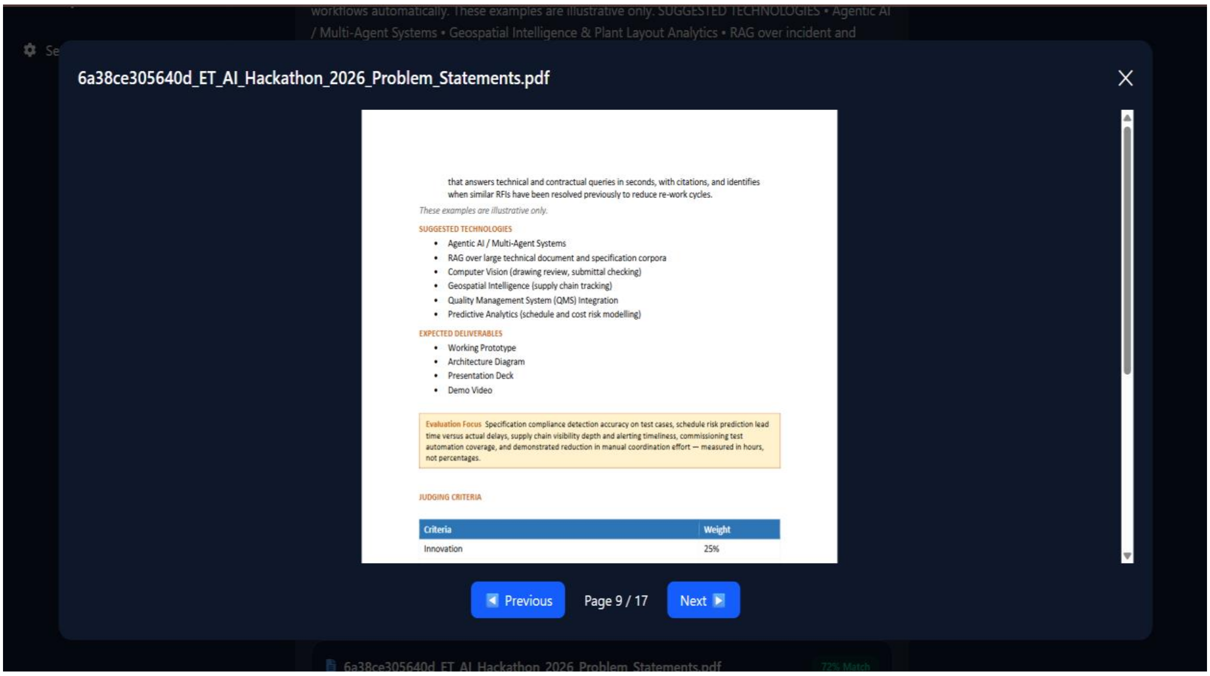
Ask a question about your uploaded knowledge base...


KnowForge AI
Industrial Knowledge Platform
Dashboard
Upload Documents
AI Assistant
Knowledge Base
Analytics
Settings
KnowForge AI Assistant
Ask questions about your uploaded knowledge base.
What technologies are suggested?
Based on the uploaded documents, I found the following relevant information:
that answers technical and contractual queries in seconds, with citations, and identifies when similar RFIs have been resolved previously to reduce re-work cycles. These examples are illustrative only. SUGGESTED TECHNOLOGIES • Agentic AI / Multi-Agent Systems • RAG over large technical document and specification corpora • Computer Vision (drawing review, submittal checking) • Geospatial Intelligence (supply chain tracking) • Quality Management System (QMS) Integration • Predictive Analy workflows automatically. These examples are illustrative only. SUGGESTED TECHNOLOGIES • Agentic AI / Multi-Agent Systems • Geospatial Intelligence & Plant Layout Analytics • RAG over incident and regulatory document corpora • Computer Vision & CCTV Analytics • IoT / SCADA Data Integration • Knowledge Graphs (equipment-permit-risk relationships) EXPECTED DELIVERABLES • Working Prototype • Architecture Diagram • Presentation Deck • Demo Video
Evaluation Focus Compound risk detect
e only. SUGGESTED TECHNOLOGIES • Geospatial Intelligence & Remote Sensing (Sentinel satellite, MODIS) • Multi-Agent AI Systems • Real-Time IoT Sensor Data Integration (CAAQMS) • Atmospheric Dispersion Modelling • Predictive Analytics • LLMs for multi-language citizen communication EXPECTED DELIVERABLES • Working Prototype • Architecture Diagram • Presentation Deck • Demo Video
Evaluation Focus Source attribution accuracy versus ground-truth emission inventories, AQI forecast acc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These examples are illustrative only.
SUGGESTED TECHNOLOGIES
Agentic AI / Multi-Agent Systems
RAG over large technical document and specification corpora
Computer Vision (drawing review, submittal checking)
Geospatial Intelligence (supply chain tracking)
Quality Management System (QMS) Integration
Predictive Analytics (schedule and cost risk modelling)
EXPECTED DELIVERABLES
Working Prototype
Architecture Diagram
Presentation Deck
Demo Video
Evaluation Focus Specification compliance detection accuracy on test cases, schedule risk prediction lead time versus actual delays, supply chain visibility depth and alerting timeliness, commissioning test automation coverage, and demonstrated reduction in manual coordination effort — measured in hours, not percentages.
JUDGING CRITERIA
Criteria
Weight
Innovation
25%
Previous
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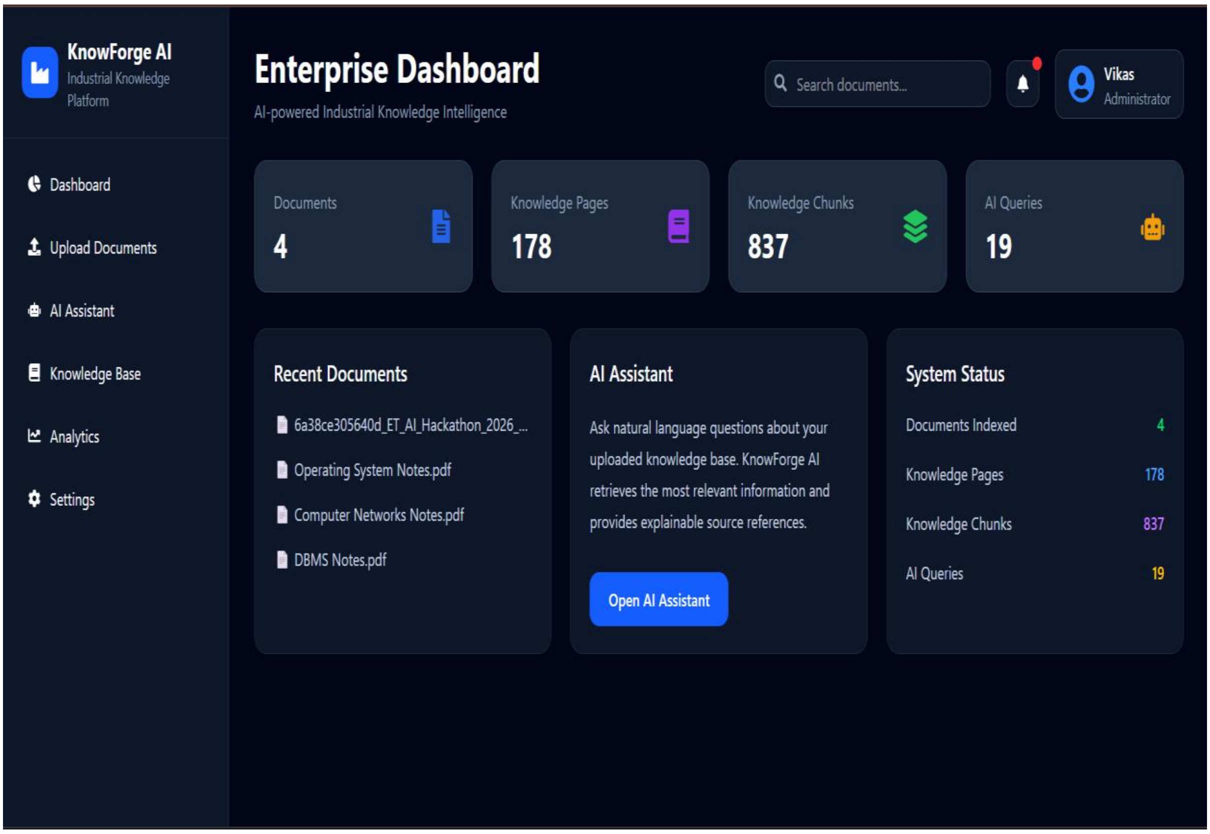
Next


KnowForge AI
Industrial Knowledge Platform
Dashboard
Upload Documents
AI Assistant
Knowledge Base
Analytics
Settings
Enterprise Dashboard
AI-powered Industrial Knowledge Intelligence
Search documents...
Vikas
Administrator
Documents
4
Knowledge Pages
178
Knowledge Chunks
837
AI Queries
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Recent Documents
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Operating System Notes.pdf
Computer Networks Notes.pdf
DBMS Notes.pdf
AI Assistant
Ask natural language questions about your uploaded knowledge base. KnowForge AI retrieves the most relevant information and provides explainable source references.
Open AI Assistant
System Status
Documents Indexed
4
Knowledge Pages
178
Knowledge Chunks
837
AI Queries
19

Future Scope

- Multi-document summarization
- OCR for scanned documents
- Role-based authentication
- Multi-language support
- Voice-based querying
- Knowledge Graph integration
- Cloud deployment
- Real-time collaboration

Conclusion

KnowForge AI successfully demonstrates how Retrieval-Augmented Generation can transform industrial knowledge management by enabling intelligent document understanding and contextual information retrieval.

The platform improves operational efficiency by allowing users to interact with industrial knowledge using natural language while maintaining explainability through source references.

The proposed solution provides a scalable foundation for future enterprise knowledge management systems.